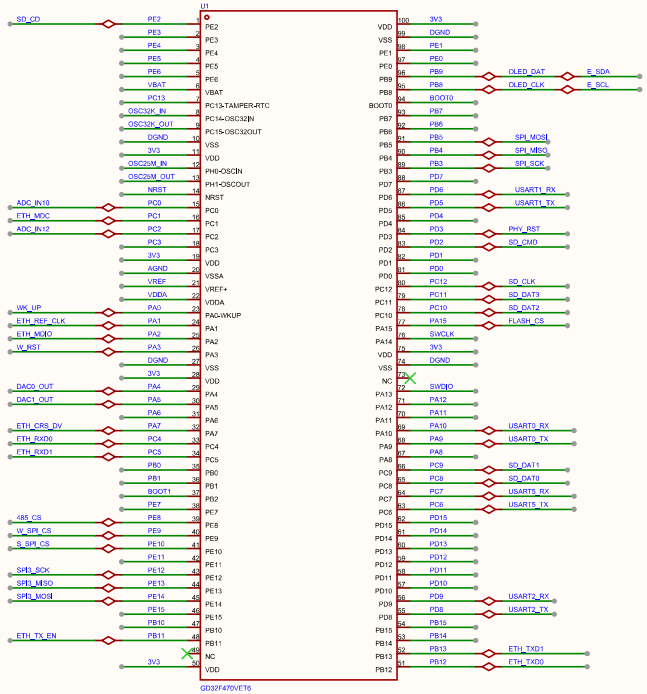
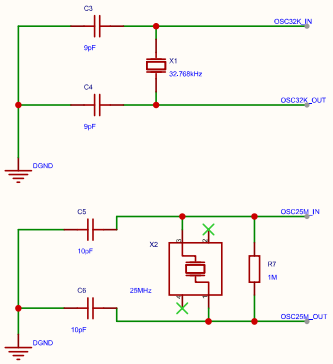
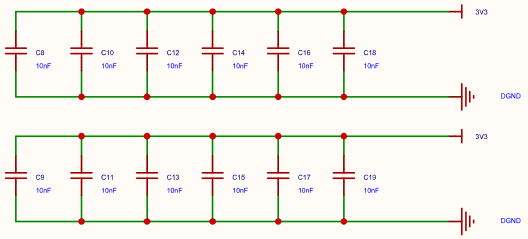



## MCU

## GD32F470VET6

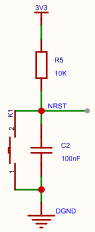



## 晶振

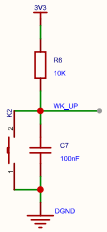



## 去耦电容

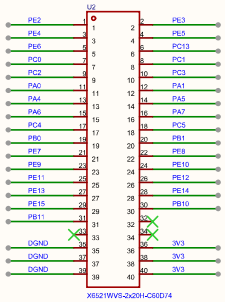
每个供电引脚旁边放俩

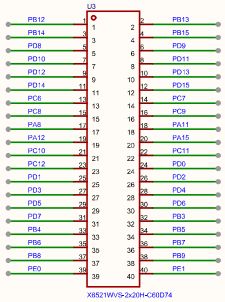


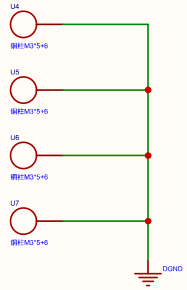

## 复位



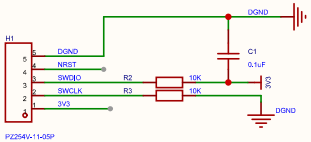

## 唤醒



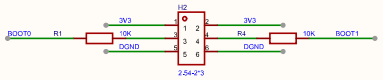




## Pin Out




SWDIO上拉电阻：抑制噪声，防止随机翻转，提升可靠性。
SWCLK下拉电阻：防止浮空导致的噪声，提升可靠性。

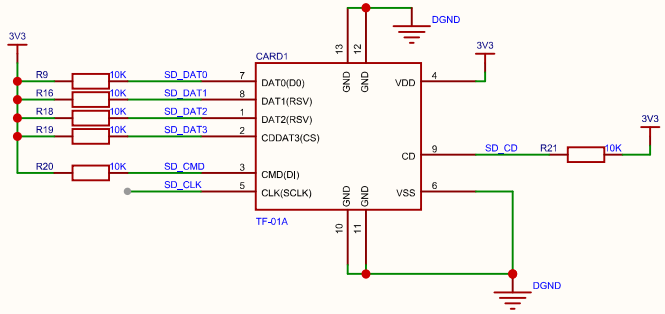
## SWD



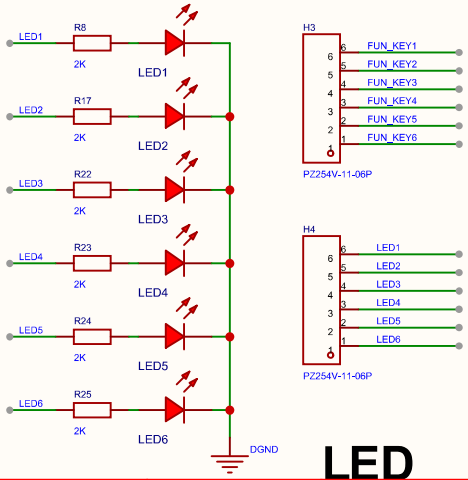

| BOOT0 | BOOT1 | 启动方式 |
|---|---|---|
| 0 | x | 内部FLASH |
| 1 | 1 | 内部SRAM |
| 1 | 0 | 系统存储器 |

## BOOT

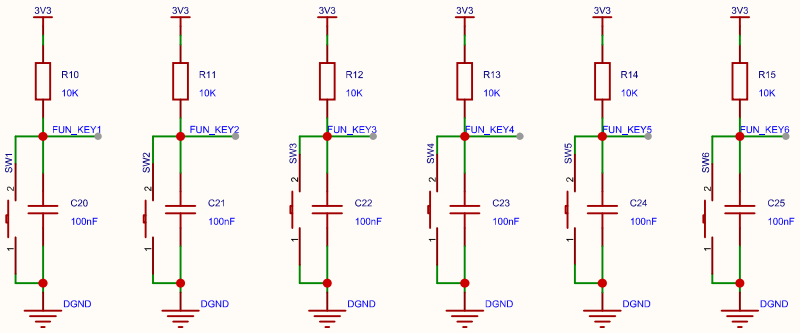
| | | | |
|---|---|---|---|
| 原理图 | CoreBoard | 创建日期 | 2026-02-06 |
| | | 更新日期 | 2026-04-07 |
| 板子 | CIMC_GD32F4_Development_Kit | 图页 | MCU_Mainboard |
| 绘制 | Lingyu Meng | GD32F470 Development Kit V2.0 | |
| 审阅 | Lingyu Meng | | |
| | | 版本 | 尺寸 | 页 1 共 5 |
| 中国智能制造挑战赛 China Intelligent Manufacturing Challenge | | V1.0 | A4 | CIMC“西门子杯”智能制造挑战赛 |



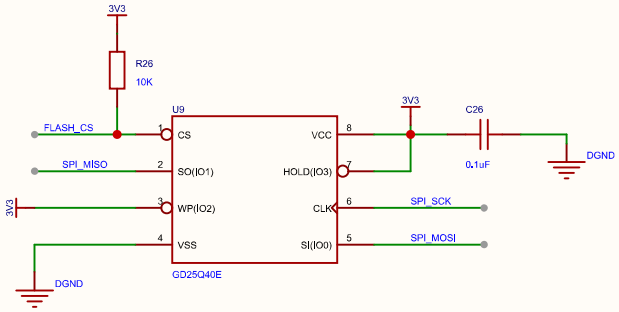

TF



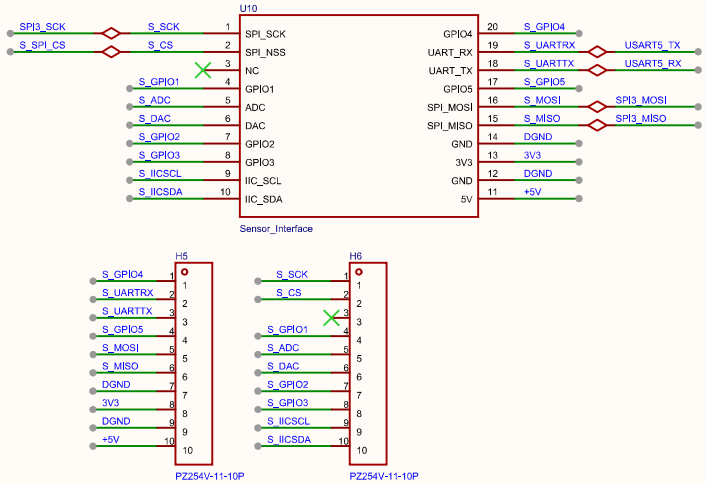

LED



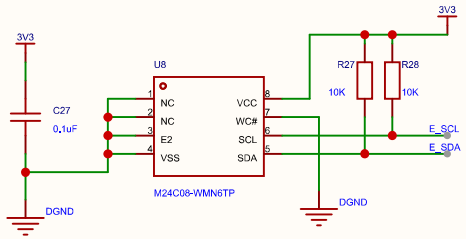

KEY



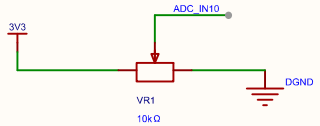

SPI FLASH



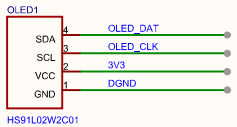

SENSOR




EEPROM



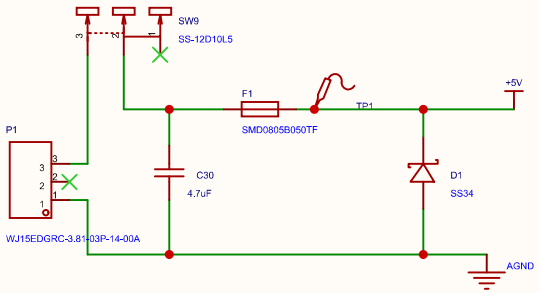

ADC

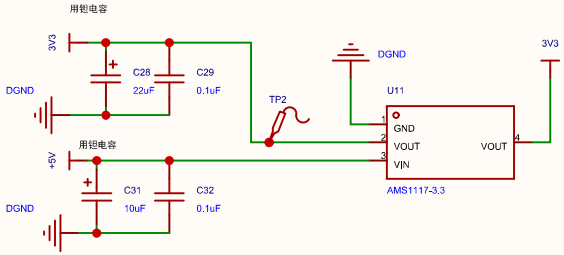



OLED

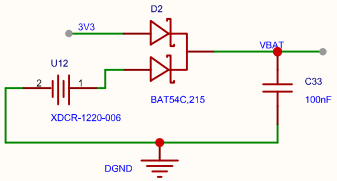
| | | | |
|---|---|---|---|
| 原理图 | CoreBoard | 创建日期 | 2026-02-06 |
| | | 更新日期 | 2026-04-07 |
| 板子 | CIMC_GD32F4_Development_Kit | 图页 | periph |
| 绘制 | Lingyu Meng | GD32F470 Development Kit V2.0 | |
| 审阅 | Lingyu Meng | | |
| | | 版本 | 尺寸 | 页 2 共 5 |
| 中国智能制造挑战赛 China Intelligent Manufacturing Challenge | | V1.0 | A4 | CIMC“西门子杯”智能制造挑战赛 |

## 5V输入

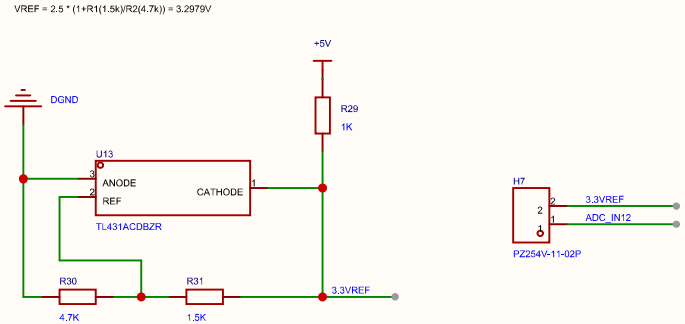


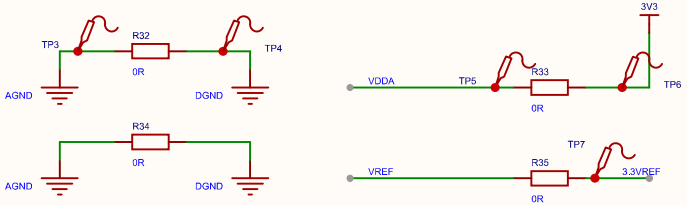

## 5V - 3V3

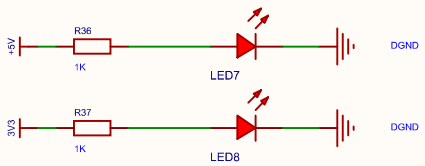



## Vbat




## ADC基准

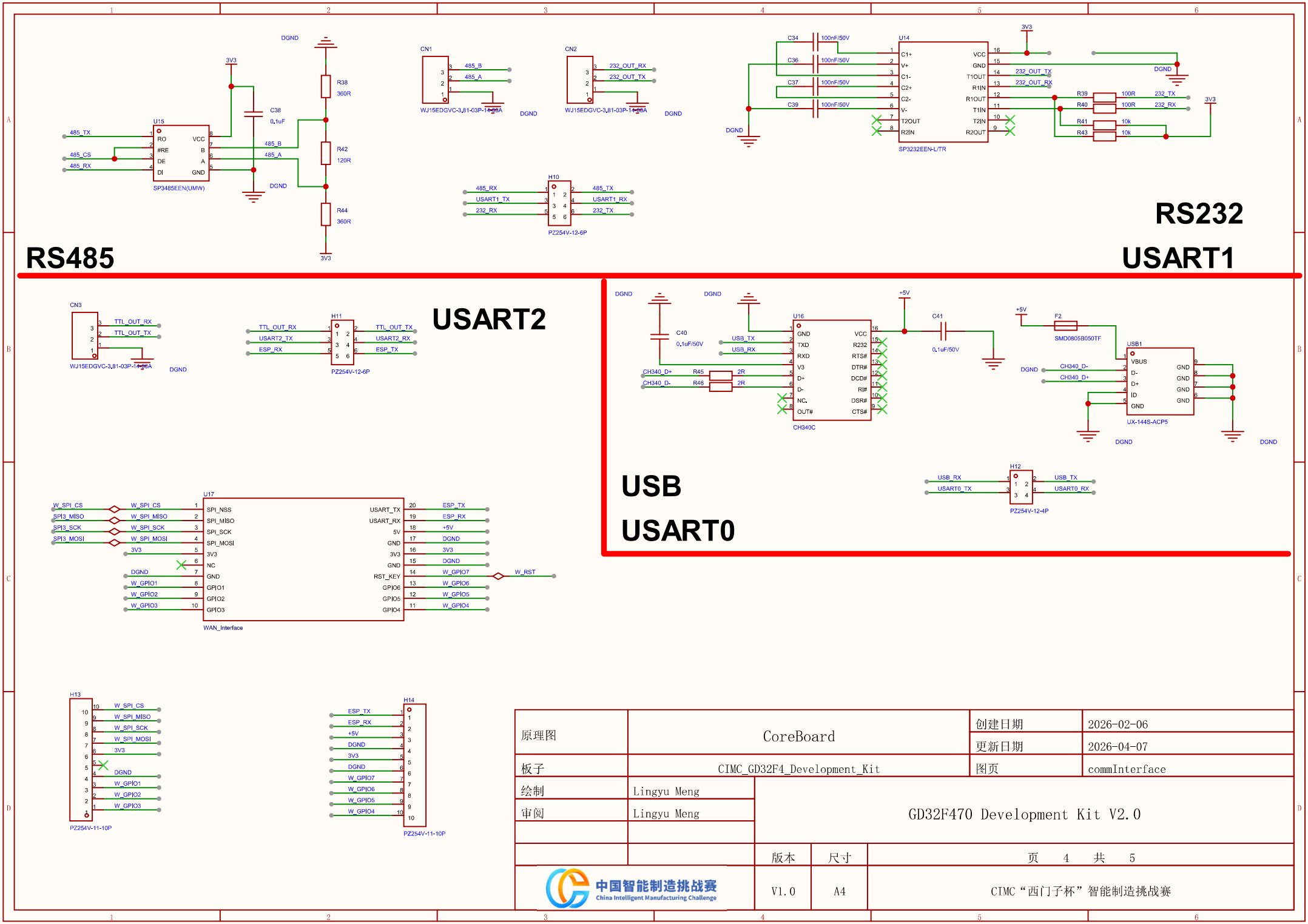




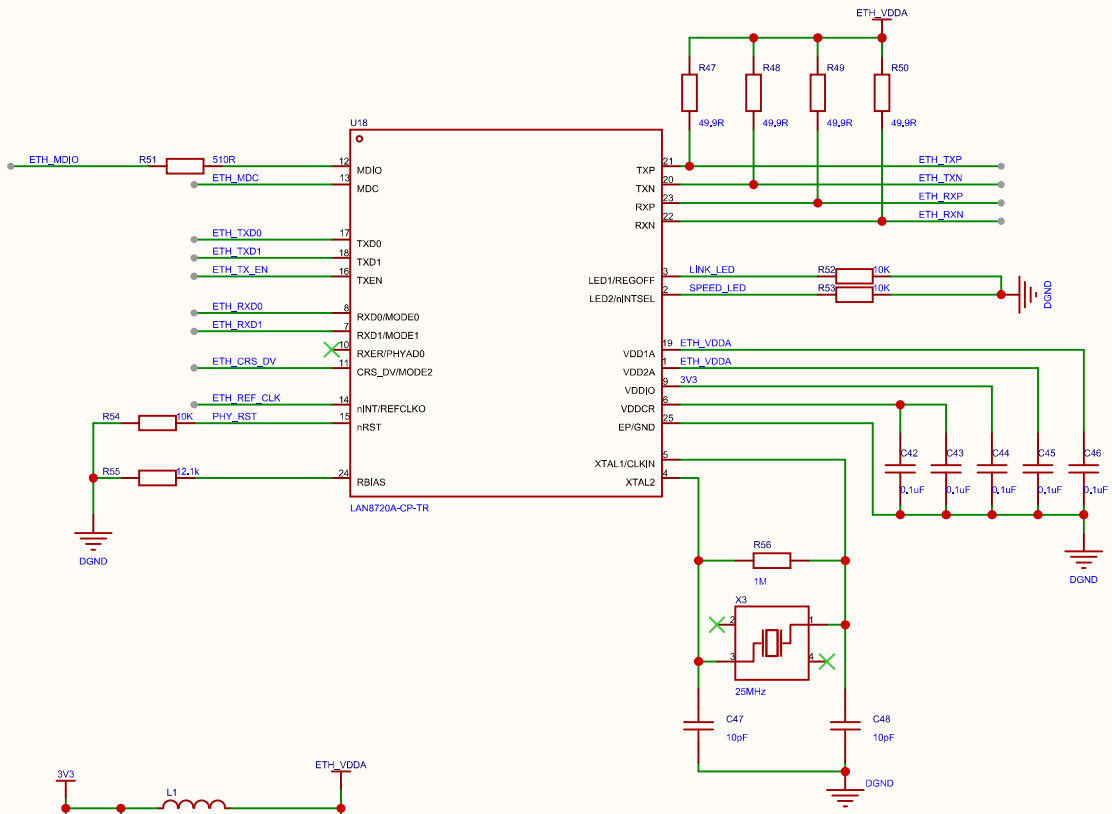

**5V 电源灯**

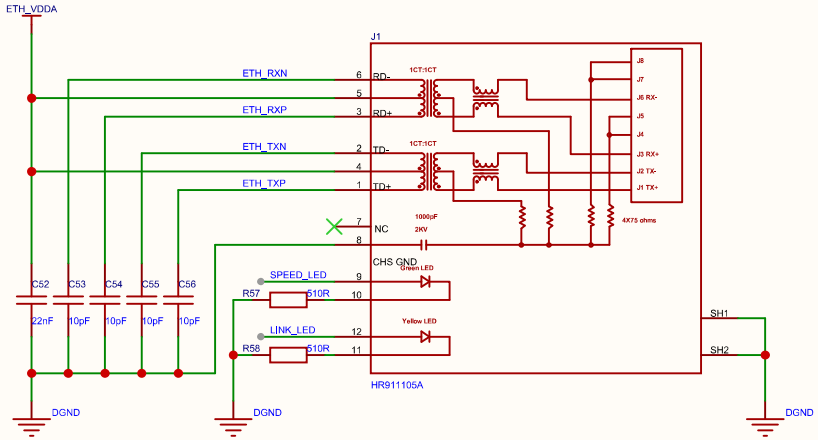
**3.3V 电源灯**

| | | | |
|---|---|---|---|
| 原理图 | CoreBoard | 创建日期 | 2026-02-06 |
| | | 更新日期 | 2026-04-07 |
| 板子 | CIMC_GD32F4_Development_Kit | 图页 | power |
| 绘制 | Lingyu Meng | GD32F470 Development Kit V2.0 | |
| 审阅 | Lingyu Meng | | |
| | | 版本 | 尺寸 | 页 3 共 5 |
| 中国智能制造挑战赛 China Intelligent Manufacturing Challenge | | V1.0 | A4 | CIMC“西门子杯”智能制造挑战赛 |




| | | | |
|---|---|---|---|
| 原理图 | CoreBoard | 创建日期 | 2026-02-06 |
| | | 更新日期 | 2026-04-07 |
| 板子 | CIMC_GD32F4_Development_Kit | 图页 | commInterface |
| 绘制 | Lingyu Meng | GD32F470 Development Kit V2.0 | |
| 审阅 | Lingyu Meng | | |
| | | 版本 | 尺寸 | 页 4 共 5 |
| 中国智能制造挑战赛 China Intelligent Manufacturing Challenge | | V1.0 | A4 | CIMC“西门子杯”智能制造挑战赛 |




| | | | |
|---|---|---|---|
| 原理图 | CoreBoard | 创建日期 | 2026-02-06 |
| | | 更新日期 | 2026-02-06 |
| 板子 | CIMC_GD32F4_Development_Kit | 图页 | Ethernet |
| 绘制 | | GD32F470 Development Kit V2.0 | |
| 审阅 | | | |
| | | 版本 | 尺寸 | 页 5 共 5 |
| 中国智能制造挑战赛 China Intelligent Manufacturing Challenge | | V1.0 | A4 | 嘉立创EDA |